

Alvaro Gomariz

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ML Scientist with 10+ years of experience in AI and ML for biomedical imaging and healthcare, including 5+ years in pharmaceutical R&D and a PhD in Computer Vision. I identify high-impact ML opportunities across research and clinical development, and translate advanced methods into reproducible, rigorously evaluated solutions. I have led cross-disciplinary projects integrating imaging and multimodal data, spanning digital pathology, microscopy, and other modalities, to support scientific and clinical decision-making.

Experience

Roche, Switzerland

Mar 2021 - Present

Senior AI/ML Scientist & Technical Lead - AI Video Colonoscopy

May 2024 - Present

- Lead the technical and scientific direction of the AI video colonoscopy program, guiding a core team of 5 AI/ML scientists and 4 contract engineers across gap analyses, rigorous model evaluation, MLOps architecture, and prioritized algorithm roadmaps, building reusable assets that can carry over to future programs
- Partner with biostatistics and clinical science to build a multimodal prognostic model (imaging + clinical data) that outperforms the clinical baseline, to be included in the statistical analysis plan of a Phase 3 trial
- Lead an annotation campaign with 5 academic experts and 4 gastroenterologists to benchmark scoring reliability and develop AI for disease-severity scoring and lesion/feature detection that matches inter-rater agreement; implement ETL pipelines and automated QC of annotations
- Curate and harmonize multimodal patient data (real-world and clinical-trial), and build an MCP interface with LLM agents that lets the team query our relational Athena database without SQL expertise
- Co-lead a data science network of 25-35 scientists and mentor interns and junior colleagues (seven interns across my Roche roles); disseminate results through conference abstracts and a patent application

AI/ML Scientist - Digital Pathology & Computational Oncology

Aug 2022 - Apr 2024

- Technically led a matrix team of 5-10 ML scientists driving a company-wide multimodal foundation model initiative on 35M+ H&E whole-slide images; adapted state-of-the-art self-supervised vision architectures (e.g., DINOv2) with distributed pre-training to deliver a proprietary model, adopted as an internal software package and disseminated in three lead-author papers
- Built a proof-of-concept for lung cancer patient stratification from H&E slides targeting signal comparable to routine genetic screening, addressing severe class imbalance and sparse signal in gigapixel images with multiple instance learning, graph neural networks, self-supervised learning, and survival analysis
- Collaborated closely with pathologists, including the project's clinical lead, to define the clinical setting, interpret whole-slide images, and assess the interpretability of model outputs
- Architected data-engineering pipelines for gigapixel-scale inputs with robust data versioning, and designed rigorous benchmarking frameworks evaluating internal models against external vendors to guide corporate build-vs-buy decisions

Postdoctoral Researcher - Ophthalmic Imaging (OCT)

Mar 2021 - Jul 2022

- Developed and published methods (semi-supervised learning, contrastive CNNs, unsupervised domain adaptation) improving ophthalmic OCT segmentation across devices, enabling predictions on new devices without additional annotations
- Prototyped and informed novel methodologies for internal OCT analysis software, extending it to new devices and real-world data, and informing subsequent foundation model efforts

MicroscopeIT, Poland & U.S.A. (remote)

Mar 2018 - Mar 2019

Technical lead

- Led R&D project from conception to delivery, creating an image analysis product leveraging deep learning and spatial statistics for large 3D microscopy clinical datasets
- Directed a team of 3-5 engineers to develop the software for biotechnology clients, including clinicians and researchers

ETH & University Hospital Zurich, Switzerland

Sept 2016 - Feb 2021

Doctoral researcher

- Developed deep learning methods (attention networks, learning with missing modalities, uncertainty estimation) for segmentation, cell detection, and spatial analysis of large multiplexed immunofluorescence and 3D microscopy datasets (confocal, light-sheet), enabling quantitative cell phenotyping across heterogeneous marker panels, leading to multiple publications in renowned scientific journals
- Collaborated with biologists, clinicians, and healthcare companies, on multiple imaging domains (MRI, ultrasound, micro-CT, and different microscopy types), including software development for users with no programming experience, leading to scientific publications and research funding
- Wrote research grants resulting in project funding, supervision of 7 students, teaching, and organization of seminars

Education

Doctor of Science in Computer Vision - ETH Zurich, Switzerland

Oct 2016 - Feb 2021

- Best PhD Award from the Center for Experimental and Clinical Imaging Technologies (EXCITE) Zurich for outstanding thesis in the field of biomedical imaging
- PhD thesis: "Quantitative Fluorescence Microscopy with Marker-Efficient and Uncertainty-Aware Segmentation and Detection Using Deep Learning"

MSc in Biomedical Engineering - ETH Zurich, Switzerland

Oct 2014 - Sep 2016

- 1st of class: awarded Willi-Studer Prize
- Master thesis: "Applying 3D quantitative microscopy to study global topography and cellular interactions in the bone marrow"
- Semester project: "Investigation of image registration for sliding boundaries"

BSc in Biomedical Engineering - University Carlos III of Madrid, Spain

Oct 2010 - Sep 2014

- 1st of class: Awarded Extraordinary End of Studies Award
- Bachelor thesis: "Resolution study of optical tomography systems in microscopy"

Skills

Machine learning	Digital pathology, foundation models, self-supervised & multiple-instance learning, GNNs / geometric deep learning, cell detection & spatial analysis, multi-modality, uncertainty estimation
Software development	MLOps, data engineering, CI/CD, ETL, high performance computing (SLURM & LSF), git
Programming & Scripting	Python, SQL, Matlab, bash/shell, Docker, R
Frameworks & tools	PyTorch, TensorFlow, AWS, V7, Nextflow, SciPy Ecosystem
Imaging tools	Fiji, Napari, 3D Slicer, ITK, OpenCV
Generative & Agentic AI	LLM APIs, MCP, LangChain, AI-assisted coding (Claude Code, Cursor)
Project leadership	Matrix teams, strategic planning, product development, cross-functional collaboration
Agile certifications	ICAgile Certified Professional (May 2024), Professional Scrum Master™ I (Aug 2022)
Languages	English (fluent), Spanish (native), German (basic)

Additional information

Scientific Contributions

JOURNAL ARTICLES

- **Gomariz A**, Kikuchi Y, Li Y, Albrecht A, Maunz A, Ferrara D, Lu H, Goksel O, “Joint semi-supervised and contrastive learning enables domain generalization and multi-domain segmentation”, *Medical Image Analysis* 103, p.103575 (2025). doi: [10.1016/j.media.2025.103575](https://doi.org/10.1016/j.media.2025.103575)
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- Vijaykumar A, Helbling PM, Thelen F, Fazio S, Mun Y, Pereira AL, Zielińska KA, Buschl P, Zerkatje T, Loosli K, Isringhausen S, Menze B, Nagasawa T, Roeder I, **Gomariz A**, Manz MG, Yokomizo T, Nombela-Arrieta C “Tissue-scale mapping reveals a central role of hepatoblasts in the regulation of fetal liver hematopoiesis and stem cell maintenance”, *Developmental Cell* 61, 901-918.e4 (2026). doi: [10.1016/j.devcel.2026.01.011](https://doi.org/10.1016/j.devcel.2026.01.011)
- **Gomariz A**, Portenier T, Nombela-Arrieta C, Goksel O, “Probabilistic Spatial Analysis in Quantitative Microscopy with Uncertainty-Aware Cell Detection using Deep Bayesian Regression”, *Science Advances* 8 (5) (2022). doi: [10.1126/sciadv.abi8295](https://doi.org/10.1126/sciadv.abi8295)
- **Gomariz A**, Portenier T, Helbling PM, Isringhausen S, Suessbier U, Nombela-Arrieta C, Goksel O, “Modality Attention and Sampling Enables Deep Learning with Heterogeneous Marker Combinations in Fluorescence Microscopy”, *Nature Machine Intelligence* p. 1-13 (2021). doi: [10.1038/s42256-021-00379-y](https://doi.org/10.1038/s42256-021-00379-y)
- Isringhausen S, Mun Y, Kovtonyuk L, Krautler N, Suessbier U, **Gomariz A**, Spaltro G, Helbling PM, Wong HC, Nagasawa T, Manz MG, Oxenius A, Nombela-Arrieta C, “Chronic viral infections persistently alter marrow stroma and impair hematopoietic stem cell fitness”, *Journal of Experimental Medicine* 218 (12) (2021). doi: [10.1084/jem.20192070](https://doi.org/10.1084/jem.20192070)
- Ghazi S, Bourgeois S, **Gomariz A**, Bugarski M, Haenni D, Martins JR, Nombela-Arrieta C, Wagner CA, Hall AM, Eilidh C, “Multiparametric imaging reveals that mitochondria rich intercalated cells in the kidney collecting duct have a very high glycolytic capacity”, *The FASEB journal* 34 (6), pp. 8510–8525 (2020). doi: [10.1096/fj.202000273R](https://doi.org/10.1096/fj.202000273R)
- **Gomariz A**, Isringhausen S, Helbling PM, and Nombela-Arrieta C, “Imaging and spatial analysis of hematopoietic stem cell niches”, *Annals of the New York Academy of Sciences* 1466(1):5-16 (2019). doi: [10.1111/nyas.14184](https://doi.org/10.1111/nyas.14184)
- Hess M, **Gomariz A**, Goksel O, Ewald CY, “In-vivo quantitative image analysis of age-related morphological changes of C. elegans neurons reveals a correlation between neurite bending and novel neurite outgrowths”, *eNeuro* 6(4) (2019). doi: [10.1523/ENEURO.0014-19.2019](https://doi.org/10.1523/ENEURO.0014-19.2019)
- Agarwal P, Isringhausen S, Li H, Paterson AJ, He J, **Gomariz A**, Nagasawa T, Nombela-Arrieta C, and Bhatia R, “Mesenchymal niche-specific expression of Cxcl12 controls quiescence of treatment-resistant leukemia stem cells”, *Cell Stem Cell* 24 (5), 769–784.e6 (2019). doi: [10.1016/j.stem.2019.02.018](https://doi.org/10.1016/j.stem.2019.02.018)
- **Gomariz A**, Helbling PM, Isringhausen S, Suessbier U, Becker A, Boss A, Nagasawa T, Paul G, Goksel O, Székely G, Stoma S, Nørrelykke SF, Manz MG, and Nombela-Arrieta C, “Quantitative spatial analysis of haematopoiesis- regulating stromal cells in the bone marrow microenvironment by 3D microscopy”, *Nature Communications* 9 (1) (2018). doi: [10.1038/s41467-018-04770-z](https://doi.org/10.1038/s41467-018-04770-z)
- Takizawa H, Fritsch K, Kovtonyuk LV, Saito Y, Yakkala C, Jacobs K, Ahuja AK, Lopes M, Hausmann A, Hardt WD, **Gomariz A**, Nombela-Arrieta C, and Manz MG, “Pathogen-Induced TLR4-TRIF Innate Immune Signaling in Hematopoietic Stem Cells Promotes Proliferation but Reduces Competitive Fitness”, *Cell Stem Cell* 21 (2), 225–240.e5 (2017). doi: [10.1016/j.stem.2017.06.013](https://doi.org/10.1016/j.stem.2017.06.013)

CONFERENCE PAPERS

- Blasco Fernandez P, Girycki R, Bossuyt P, Daperno M, Kopylov U, Bouhnik Y, Karmiris K, Stimpel B, Levitte S, **Gomariz A** “Evaluating the Impact of Temporal Video Context on the Assessment of Endoscopic Scoring in Ulcerative Colitis”, *United European Gastroenterology Journal*, UEG Week 2026 (accepted)
- Blasco Fernandez P, Girycki R, Bossuyt P, Daperno M, Kopylov U, Bouhnik Y, Karmiris K, Stimpel B, Levitte S, **Gomariz A** “Characterizing the Reliability and Scalability of Granular Mucosal Labeling in Ulcerative Colitis Across Rater Groups”, *United European Gastroenterology Journal*, UEG Week 2026 (accepted)
- Blasco Fernandez P, Bossuyt P, Daperno M, Kopylov U, Bouhnik Y, Karmiris K, Benmansour F, Gutierrez-Becker B, Fraessle S, Stimpel B, Levitte S, **Gomariz A** “Quantifying the Spatial Inter-Rater Reliability of Mucosal Lesion Annotations in Ulcerative Colitis”, *Gastrointestinal Endoscopy* 103(5), pp. S-550 (2026)
- Blasco Fernandez P, Bossuyt P, Daperno M, Kopylov U, Bouhnik Y, Karmiris K, Benmansour F, Gutierrez-Becker B, Fraessle S, Stimpel B, Levitte S, **Gomariz A** “Quantifying Inter-Rater Reliability of Mucosal Features in Ulcerative Colitis Endoscopy”, *Journal of Crohn's and Colitis* Volume 20, Issue Supplement_1 (2026). doi: [10.1093/ecco-jcc/jjaf231.052](https://doi.org/10.1093/ecco-jcc/jjaf231.052)
- **Gomariz A**, Gutierrez-Becker B, Girycki R, Szostak P, Stimpel B, Levitte S, Abbasi-Sureshjani S “Assessing the Impact of Biopsy Artifacts on AI-Based Endoscopic Scoring for Ulcerative Colitis”, *Journal of Crohn's and Colitis* Volume 20, Issue Supplement_1 (2026). doi: [10.1093/ecco-jcc/jjaf231.699](https://doi.org/10.1093/ecco-jcc/jjaf231.699)
- **Gomariz A**, Szostak P, Girycki R, Stimpel B, Abbasi-Sureshjani S, Levitte S, Karrer T “Leveraging Global Video Context to Automate Colon Section Identification in Ulcerative Colitis Endoscopy”, *United European Gastroenterology Journal* Volume 13, Issue S8 (2025)
- **Gomariz A**, Stimpel B, Stefanescu C, Gutierrez-Becker B, Frässle S, Girycki R, Fisher E, Bouhnik Y, Levitte S, Karrer T “Real-World Evidence Validates AI-Based Endoscopic Scoring for Ulcerative Colitis”, *United European Gastroenterology Journal* Volume 13, Issue S8 (2025)
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- Bredell G, Fischer M, Szostak P, Abbasi-Sureshjani S, and **Gomariz A**, “The Importance of Downstream Networks in Digital Pathology Foundation Models”, *International Workshop on Foundation Models for General Medical AI - MICCAI* pp. 10-19 (2024). doi: [10.1007/978-3-031-73471-7_2](https://doi.org/10.1007/978-3-031-73471-7_2)
- Ibañez V, Szostak P, Wong Q, Korski K, Abbasi-Sureshjani S, and **Gomariz A**, “Integrating multiscale topology in digital pathology with pyramidal graph convolutional networks”, *IEEE International Symposium on Biomedical Imaging (ISBI)* pp. 1-4 (2024). doi: [10.1109/ISBI56570.2024.10635702](https://doi.org/10.1109/ISBI56570.2024.10635702)
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- Li Y, Lu H, Albrecht T, Maunz A, Neubert A, Benmansour F, **Gomariz A**, Luu J, Ferrara D “Deep learning (DL) to segment retinal layer disruption on optical coherence tomography (OCT) in neovascular age-related macular degeneration (nAMD)”, *Investigative Ophthalmology & Visual Science, ARVO abstract* (2022)
- **Gomariz A**, Lu H, Li Y, Albrecht A, Maunz A, Benmansour F, Luu J, Goksel O, Ferrara D, “A unified deep learning approach for OCT segmentation from different devices and retinal diseases”, *Investigative Ophthalmology & Visual Science, ARVO abstract* (2022)
- **Gomariz A**, Egli R, Portenier T, Nombela-Arrieta C, and Goksel O, “Utilizing Uncertainty Estimation in Deep Learning Segmentation of Fluorescence Microscopy Images with Missing Markers”, *IEEE International Symposium on Biomedical Imaging (ISBI)* pp. 371-374 (2021). doi: [10.1109/ISBI48211.2021.9434158](https://doi.org/10.1109/ISBI48211.2021.9434158)

- **Gomariz A**, Li W, Ozkan E, Tanner C, and Goksel O, “Siamese Networks with Location Prior for Landmark Tracking in Liver Ultrasound Sequences”, *IEEE International Symposium on Biomedical Imaging (ISBI)* pp. 1757-1760 (2019). doi: [10.1109/ISBI.2019.8759382](https://doi.org/10.1109/ISBI.2019.8759382)
- Goksel O, Vishnevsky V, **Gomariz A**, and Tanner C, “Imaging of Sliding Visceral Interfaces During Breathing”, *IEEE International Symposium on Biomedical Imaging (ISBI)* pp. 298-301 (2016). doi: [10.1109/ISBI.2016.7493268](https://doi.org/10.1109/ISBI.2016.7493268)